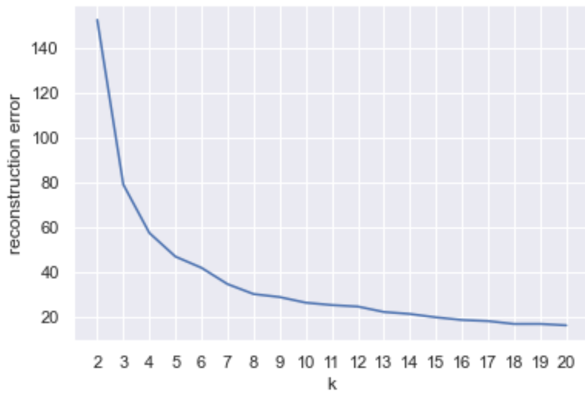
```
Part1.
Iteration 0: finds better clustering, latest reconstruction error is: 78.8
5566582597727
Iteration 2: finds better clustering, latest reconstruction error is: 78.8
51441426146
Score diff smaller than 1%, stop.
```

plot reconstruction error with k:




```
Manually found elbow: 6
```

Self-implemented algorithm to find it programmatically:

```
Try to do it algorithmically by calculating the speed of reconstruction er
ror decline
Finding elbow k programatically...
Decline ratios calculated.
inspecting no.0 in raios
inspecting no.1 in raios
inspecting no.2 in raios
inspecting no.3 in raios
valid ratio counts inside window: 2, less than a half. Stop!
Found the elbow idx in ratios: 5, transforming to k...
The elbow k found: 8

Cannot calculate Accuracy Score because the number of classes is not the s
ame as the number of clusters

confusion matrix:
```

	cluster0	cluster1	cluster2	cluster3	cluster4	cluster5
setosa	0	22	0	0	0	28
versicolor	0	0	26	0	24	0
virginica	24	0	13	12	1	0

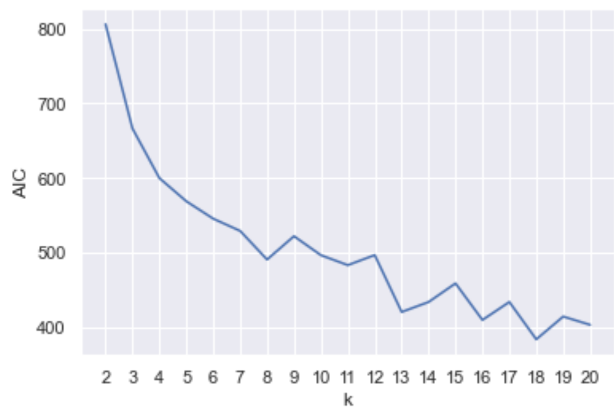
Classify the entire iris data set using cluster k=3

```
Accuracy score: 0.8933333333333333
confusion matrix:
```

	cluster0	cluster1	cluster2
setosa	0	50	0
versicolor	48	0	2
virginica	14	0	36

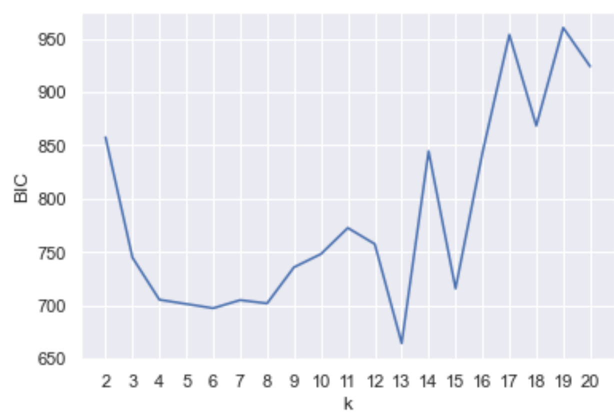
```
Part2.
Iteration 0: found better lower_bound_ score: -1.2014746139064725
Iteration 44: found better lower_bound_ score: -1.2014746139064734
Score diff smaller than 1%, stop.
```

Plot AIC with k:




```
The elbow of the curve (aic_elbow_k): 8
```

Plot BIC with k:




```
The elbow of the curve (bic_elbow_k): 4
```

Classify the entire iris data set using aic_elbow_k=8

```
Cannot calculate Accuracy Score because the number of classes is not the same
 as the number of clusters
confusion matrix:
             cluster0  cluster1  cluster2  cluster3  cluster4  cluster5  \
setosa             15         0         0         0         0        35
versicolor          0         9         1        21         4         0
virginica           0        12        25         1         0         0

            cluster6  cluster7
setosa             0         0
versicolor        15         0
virginica          0        12
```

Classify the entire iris data set using bic_elbow_k=4

```
Cannot calculate Accuracy Score because the number of classes is not the s
ame as the number of clusters.
confusion matrix:
            cluster0  cluster1  cluster2  cluster3
setosa             0        50         0         0
versicolor        33         0         1        16
virginica          0         0        45         5
```

Classify the entire iris data set using k=3

```
Accuracy score: 0.9666666666666667
confusion matrix:
            cluster0  cluster1  cluster2
setosa             0        50         0
versicolor        45         0         5
virginica          0         0        50
```